

Victor Garcia

Altermagnets

Metamaterial
setup

Two units

Chain

Lieb lattice

Conclusions

Synchronization in mechanical altermagnet

Victor Garcia

University of Amsterdam

2025-2026



Magnetism

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setup

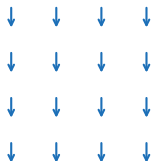
Two units

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Conclusions

- Magnets break spin-rotational/time-reversal symmetry



Ferromagnet

Magnetism

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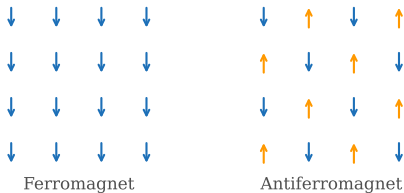
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Conclusions

- Magnets break spin-rotational/time-reversal symmetry



- Search for real space transformation that combined with time-reversal becomes a symmetry

Magnetism

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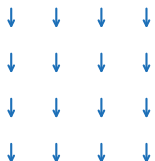
Two units

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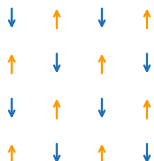
Lieb lattice

Conclusions

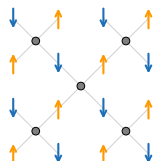
- Magnets break spin-rotational/time-reversal symmetry



Ferromagnet



Antiferromagnet



Altermagnet

- Search for real space transformation that combined with time-reversal becomes a symmetry
- Discovery of altermagnets in 2022¹

¹ L. Smejkal, J. Sinova, and T. Jungwirth. "Emerging research landscape of altermagnetism". In: *Physical Review X* 12 (2022), p. 040501

Magnets for storing data

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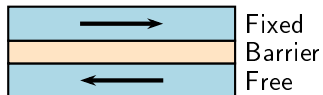
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Conclusions



Bit = 0



Bit = 1

Magnets for storing data

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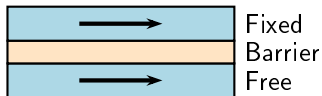
Metamaterial
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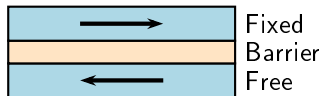
Chain

Lieb lattice

Conclusions



Bit = 0



Bit = 1

Ferromagnets:

Pro Large spin-splitting

Con $M \neq 0$

Antiferromagnets:

Con No spin-splitting

Pro $M = 0$

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Altermagnetic symmetry

Real space rotation + spin-flip denoted by $C_n\mathcal{T}$

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Altermagnetic symmetry

Real space rotation + spin-flip denoted by $C_n\mathcal{T}$

Properties

- 1 No net magnetization $M = 0$
- 2 Spin-split bands $E_{\uparrow} \neq E_{\downarrow}$
- 3 **Spin-momentum locking** $\vec{v}_{\uparrow} \neq \vec{v}_{\downarrow}$

Found in nature: MnTe, CrSb, RuO₂

Lieb lattice as a model altermagnet

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Lieb lattice as a model altermagnet

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Altermagnets

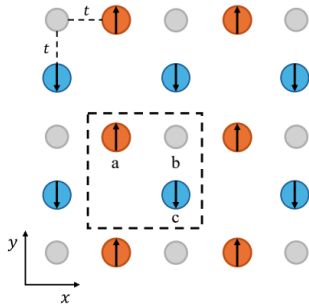
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Conclusions



- 2d square lattice with 3-atom basis
- Symmetry: $C_4\mathcal{T}$
- Spin-momentum locking

visser2023

Generalization of altermagnetic symmetry

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Conclusions

Generalization $C_4\mathcal{T}$ symmetry

spin $\longrightarrow$ 2-state variable

Generalization of altermagnetic symmetry

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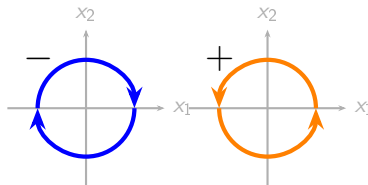
Conclusions

Generalization $C_4\mathcal{T}$ symmetry

spin $\rightarrow$ 2-state variable

Some examples are:

- Photon polarization $\rightarrow$ photonic altermagnet¹
- Electric dipole $\rightarrow$ alterelectric²
- Oscillator chirality $\rightarrow$ mechanical altermagnet



¹ Andrew Sungwook Kim et al. *Photonic Altermagnets: Magnetic Symmetries in Photonic Structures*. 2025. arXiv: 2506.23497 [physics.optics]. URL: <https://arxiv.org/abs/2506.23497>

² visser2023

Experimental setup

- How to build a mechanical altermagnet?

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Conclusions

Experimental setup

- How to build a mechanical altermagnet? →
Metamaterials

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Conclusions

- How to build a mechanical altermagnet? →
Metamaterials
- How to implement chirality?

Experimental setup

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**Metamaterial
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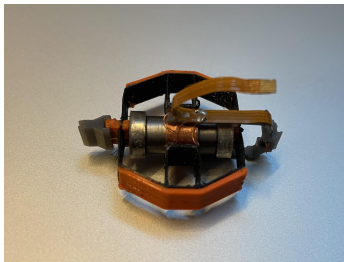
Two units

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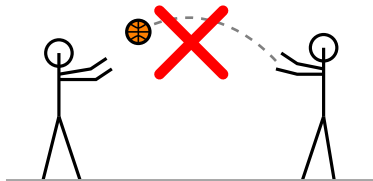
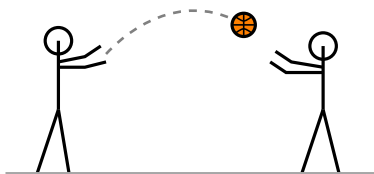
Lieb lattice

Conclusions

- How to build a mechanical altermagnet? →
Metamaterials
- How to implement chirality?
→ Non-reciprocity between two linear actuators

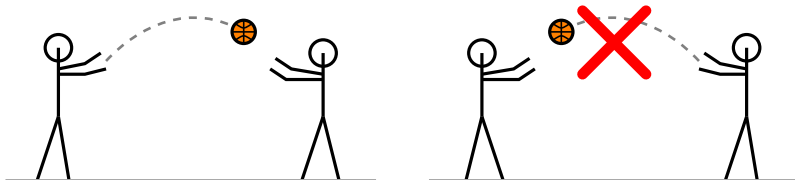


Non-reciprocity



- Asymmetry in interaction

Non-reciprocity



- Asymmetry in interaction
- Odd elasticity: $F_1 = k_a x_2$ $F_2 = -k_a x_1$

Experimental control parameter

Activity/gain k_a

- No conservation of energy
 - 1 Active
 - 2 Non-hermitian

Non-reciprocity leads to limit cycles

EOM of two non-reciprocally coupled actuators:

$$\ddot{x}_1 + k_0 x_1 + \gamma \dot{x}_1 + k_c x_1^3 + k_a x_2 = 0$$

$$\ddot{x}_2 + k_0 x_2 + \gamma \dot{x}_2 + k_c x_2^3 - k_a x_1 = 0$$

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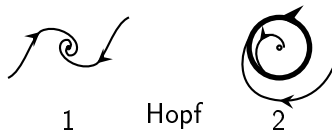
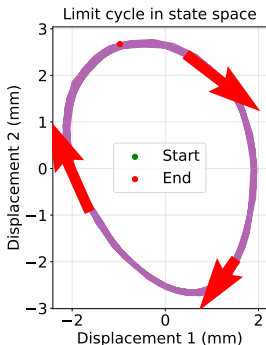
Conclusions

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$$\ddot{x}_2 + k_0 x_2 + \gamma \dot{x}_2 + k_c x_2^3 - k_a x_1 = 0$$



- 1 If $|k_a| < \gamma\sqrt{k_0}$, oscillations die
- 2 If $|k_a| > \gamma\sqrt{k_0}$, stable limit cycles, with chirality
 - a $k_a > 0$ counterclockwise
 - b $k_a < 0$ clockwise

Experimental results of one unit

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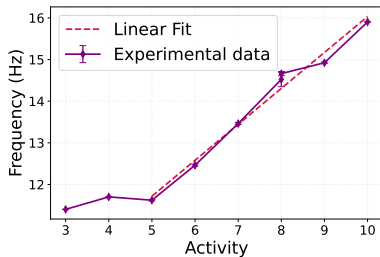
Metamaterial
setup

Two units

Chain

Lieb lattice

Conclusions



Theory predicts: $\omega = \frac{k_a}{\gamma}$

Experimental results of one unit

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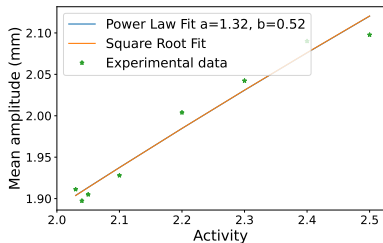
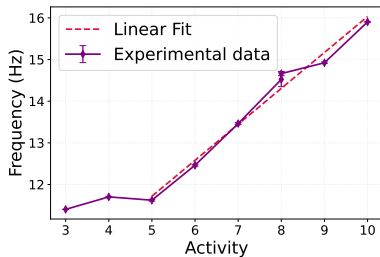
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Conclusions



Theory predicts: $\omega = \frac{k_a}{\gamma}$

$$R \sim \sqrt{\frac{2k_a^c}{k_c \gamma^2} (k_a - k_a^c)}$$

Experimental setup

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setup**

Two units

Chain

Lieb lattice

Conclusions

Next: adding more units

- Neighbour interactions through elastic band

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setup

Two units

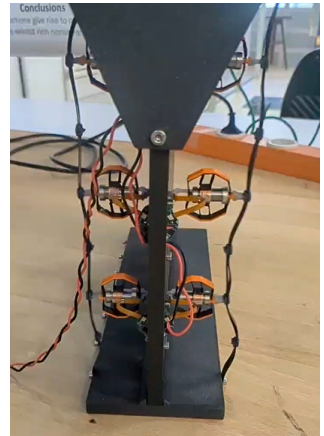
Chain

Lieb lattice

Conclusions

Next: adding more units

- Neighbour interactions through elastic band
- Different geometries
 - 1 One unit: two actuators
 - 2 Two units
 - 3 1d chain
 - 4 Cross cell
 - 5 Lieb lattice



Synchronization

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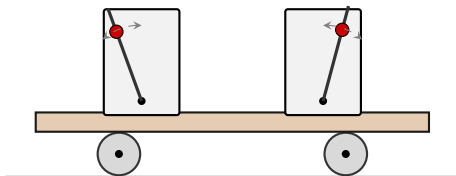
**Metamaterial
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Two units

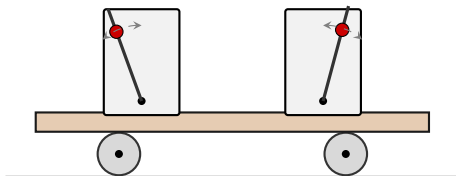
Chain

Lieb lattice

Conclusions



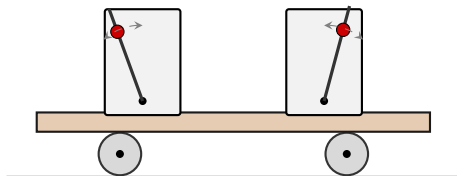
Synchronization



Requirements:

- Autonomous oscillators
- Interactions: via elastic band $\rightarrow$ weak
- Parameter proximity

Synchronization



Requirements:

- Autonomous oscillators
- Interactions: via elastic band $\longrightarrow$ weak
- Parameter proximity

Adler equation:

$$\frac{d\Delta\phi}{dt} = \underbrace{\Delta\omega}_{\omega_1 - \omega_2} - K \sin(\Delta\phi)$$

Synchronization

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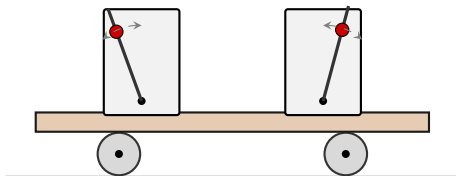
Metamaterial
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Conclusions



Requirements:

- Autonomous oscillators
- Interactions: via elastic band $\rightarrow$ weak
- Parameter proximity

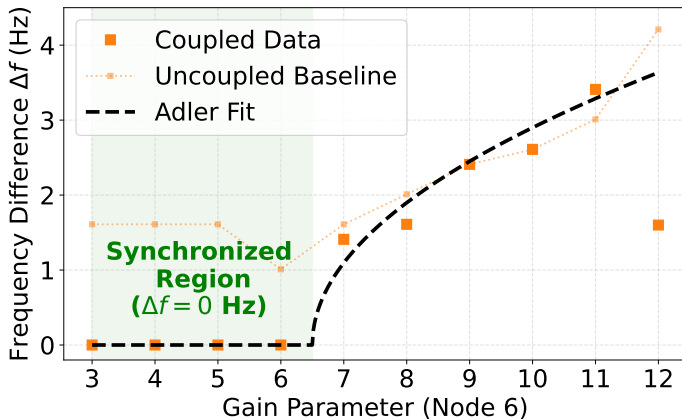
Adler equation:

$$\frac{d\Delta\phi}{dt} = \underbrace{\Delta\omega}_{\omega_1 - \omega_2} - K \sin(\Delta\phi)$$

$\Delta\omega < K$ Synchronization

$\Delta\omega > K$ Chaos

Two units with same chirality synchronize



If activity difference or detuning is large,
they stop synchronizing



Units with opposite chirality do not synchronize

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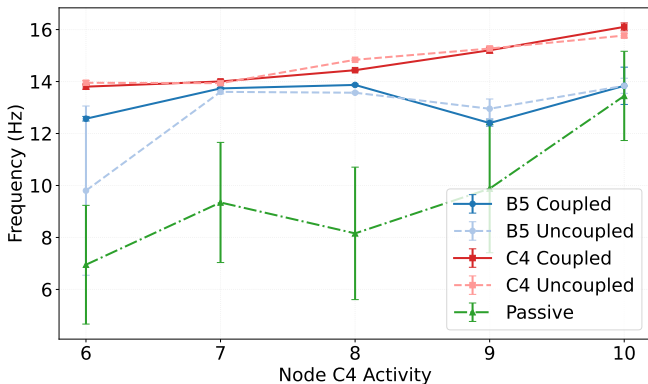
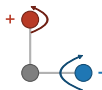
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Conclusions



Passive unit introduces an extra mode that absorbs energy

1d chain with same chirality synchronizes

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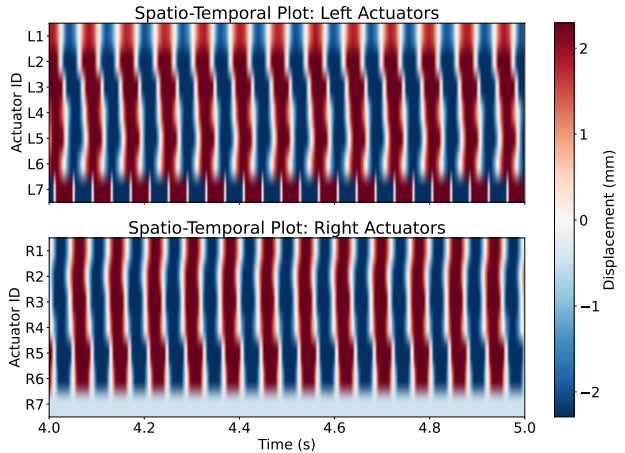
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Conclusions



Synchronization instead of wave propagation due to autonomous oscillators

1d chain with opposite chirality attenuates I

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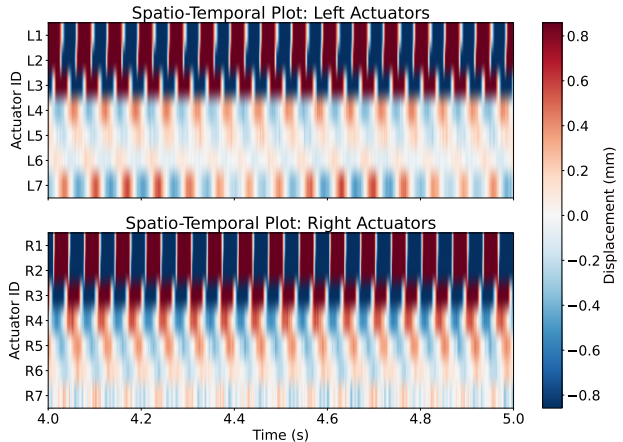
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Conclusions



Oscillation amplitude is significantly reduced

1d chain with opposite chirality attenuates II

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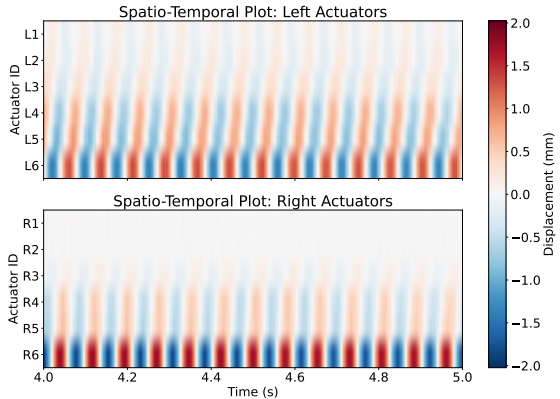
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Conclusions



Spin-momentum locking

Long chain exhibits wave coarsening

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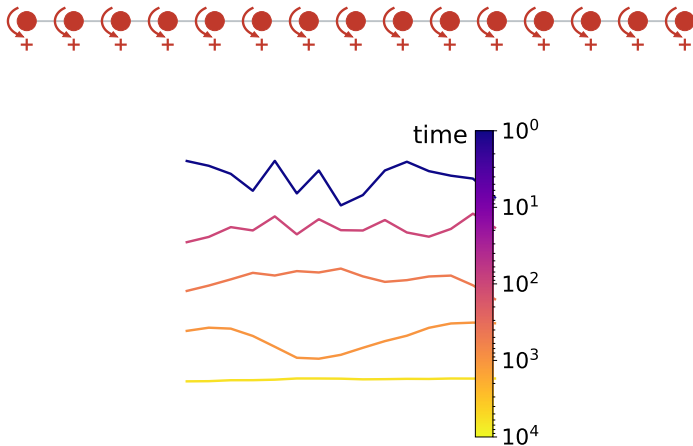
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Conclusions



Simulation of 15 units

Frustration in cross cell

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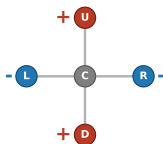
Metamaterial
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Two units

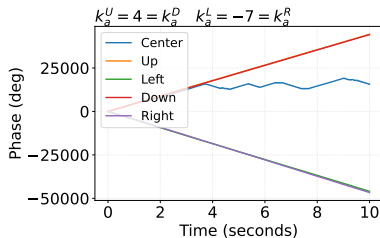
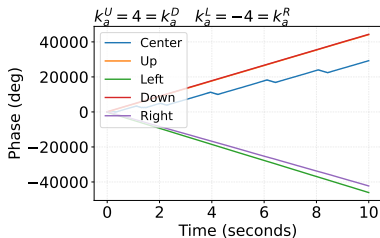
Chain

Lieb lattice

Conclusions

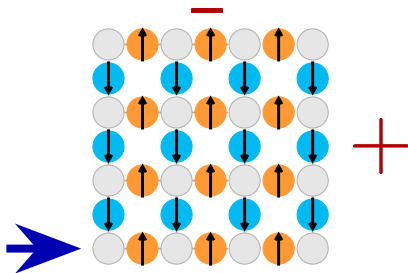


The passive center unit is
bistable



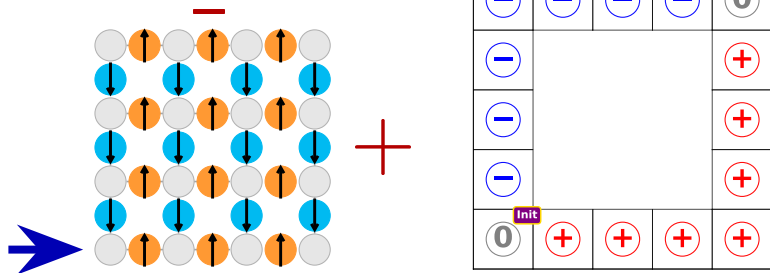
Mechanical Lieb lattice

Below Hopf bifurcation: spin-momentum locking



Mechanical Lieb lattice

Below Hopf bifurcation: spin-momentum locking



Chirality measured as the sign of $J = \dot{x}_1 x_2 - x_1 \dot{x}_2$

Chirality in mechanical Lieb lattice

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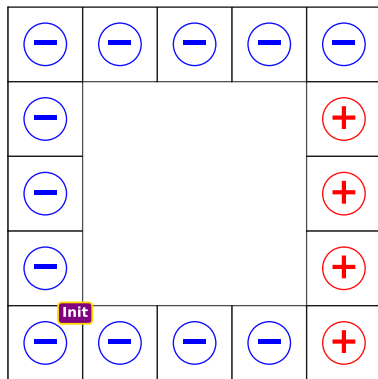
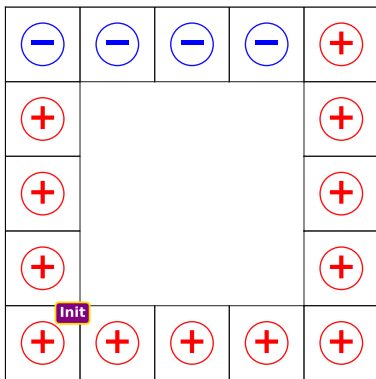
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Chirality measured as the sign of $J = \dot{x}_1 x_2 - x_1 \dot{x}_2$

Conclusions

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Conclusions

- Implemented "mechanical spin"

Conclusions

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Conclusions

- Implemented "mechanical spin"
- Built a mechanical altermagnet

Conclusions

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- Implemented "mechanical spin"
- Built a mechanical altermagnet

Main conclusion

Spin-momentum locking $\longrightarrow$ selective propagation in different directions

Acknowledgements

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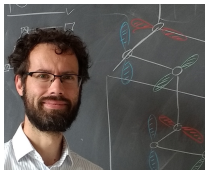
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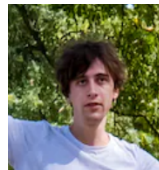
Jasper van
Wezel



Corentin
Coulais



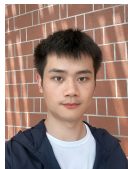
Ronald Hassing



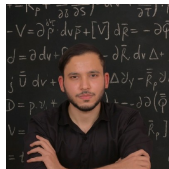
Kasper van
Nieuwland



Viktor Könye



Yuan Zhou



Rupesh Mahore



Mels van Eck